(22) $y = \sin^3 x + \cos^3 x$ (23) $y = \sin^2(\cos kx)$ (24) $y = (1+\cos^2 x)^6$

(25) $y = \dfrac{e^{3x}}{1+e^x}$ (26) $y = \sqrt{x+\sqrt{x}}$ (27) $y = e^{x\cos x}$

(28) $y = \sqrt{x+\sqrt{x+\sqrt{x}}}$ (29) $y = \sin\left(\tan\left(\sqrt{\sin x}\right)\right)$ (30) $y = \sin^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$

10.4.3 Implicit Differentiation

A function in which the dependent variable is expressed solely in terms of the independent variable x, namely, $y = f(x)$, is said to be an **explicit** function. For instance, $y = \frac{1}{2}x^3 - 1$ is an explicit function, whereas an equivalent equation $2y - x^3 + 2 = 0$ is said to define the function **implicitly** or y is an **implicit function** of x.

Now, as we know, the equation

$$x^2 + y^2 = 4 \tag{1}$$

describes *a* a circle of radius 2 centered at the origin. Equation (1) is not a function since for any choice of x in the interval $-2 < x < 2$ there correspond two values of y, namely

$$f(x) = \sqrt{4-x^2}, -2 \le x \le 2 \tag{2}$$

$$g(x) = -\sqrt{4-x^2}, -2 \le x \le 2 \tag{3}$$

(2) represents the top half of the circle (1) and (3) represents the bottom half of the circle (1). By considering either the top half or bottom half, of the circle, we obtain a function. We say that (1) defines at least two implicit functions of x on the interval $-2 \le x \le 2$.

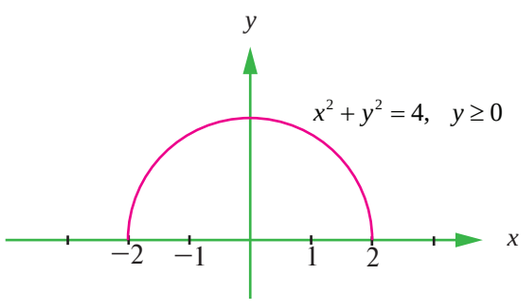


Fig. 10.25

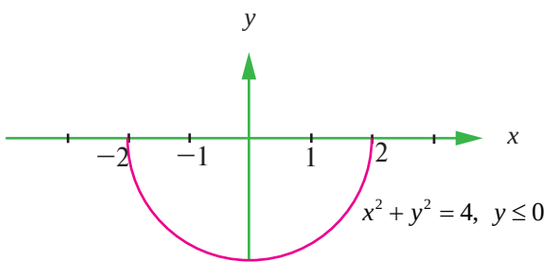


Fig. 10.26

Note that both equations

$x^2 + [f(x)]^2 = 4$ and $x^2 + [g(x)]^2 = 4$ are identities on the interval $-2 \le x \le 2$.

In general, if an equation $F(x, y) = 0$ defines a function f implicitly on some interval, then $F(x, f(x)) = 0$ is an identity on the interval. The graph of f is a portion (or all) of the graph of the equation $F(x, y) = 0$.

A more complicated equation such as $x^4 + x^2y^3 - y^5 = 2x + 1$ may determine several implicit functions on a suitably restricted interval of the x-axis and yet it may not be possible to solve for y in terms of x. However, in some cases we can determine the derivative $\dfrac{dy}{dx}$ by a process known

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